

Healthcare Automation & Chatbot System

End-to-End WhatsApp & Web Solution for Home Healthcare, Labs & Diagnostic Services

CLIENT / LEAD

AMPLR Health

DOCUMENT PURPOSE

Architecture Proposal

RECOMMENDED TECH STACK

Supabase + Node.js + Meta
API

PREPARED BY

AutomateX.ai Engineering

1 Executive Lead Overview (AMPLR Health Requirement)

Context: A high-intent lead was captured via AutomateX website for *AMPLR Health (Location: AP)* requesting an automated healthcare chatbot for doorstep medical services.

CLIENT CONTACT

Ranga Muni (AMPLR Health)

INDUSTRY

Home Healthcare & Diagnostics

SERVICES REQUESTED

Nursing, Caretaker, Physio,
Lab, ECG

2 End-to-End Chatbot Automation Flow

Unlike a simple FAQ bot, this solution automates the complete service lifecycle from initial inquiry to booking, staff allocation, payment, and post-visit feedback.

1 Welcome & Language Selection

Interactive WhatsApp/Web greeting with instant service options (Nursing, Caretaker, Physio, Lab Test, ECG, Existing Booking).

2 Requirement Capture Engine

Gathers essential patient data: Patient Name, Age, Pincode/Location, Preferred Date & Time Slot, Duration, and Medical Notes.

3 Dynamic Availability & Quotation Engine

Checks available health professionals for the requested area & slot. Automatically calculates service pricing and displays instant quotation.

4 Instant Booking & Payment Link

Generates booking ID, creates record in database, and sends automated Razorpay payment link via WhatsApp.

5 Staff Assignment & Visit Status Pipeline

Allocates nearest available nurse/physio. Updates status in real-time: *Assigned* → *Accepted* → *On the Way* → *In Progress* → *Completed*.

6 Invoice, Feedback & Re-Booking Automation

Sends PDF receipt after completion, collects star rating/feedback, and provides 1-click "Rebook Same Service" shortcut.

Healthcare AI Safety Layer: Clinical / Emergency Escalation Branch

Critical Safety Rule: The bot is strictly programmed NOT to provide medical diagnosis. If the patient inputs critical emergency triggers, the automated booking flow pauses immediately, an Emergency Alert message is shown, and the case is escalated to the human clinical response team instantly.

 Severe Chest Pain

 Breathing Difficulty

 Unconscious Patient

 Heavy Bleeding

 Sudden Trauma / Emergency

3 Backend Choice: Can We Use Supabase? (Detailed Analysis)

⚡ Recommendation: YES! Supabase is the PERFECT Backend Choice

HIGHLY RECOMMENDED

Why Supabase instead of buying an external heavy CRM?

Building the backend on **Supabase (PostgreSQL + Realtime Engine + Auth + Edge Functions)** provides 100% control over patient data, zero monthly enterprise licensing costs, instant real-time status sync for the Admin Dashboard, and extreme developer speed.

1. Real-time Status Sync (WebSockets)

When a nurse accepts a job or updates status ("On the way"), Supabase Realtime pushes live updates to the Admin Dashboard without refreshing.

2. PostgreSQL + Row-Level Security

Enterprise-grade medical data security. Patient records, medical notes, and payment histories are strictly isolated per role.

3. Supabase Edge Functions

Serverless TypeScript functions handle WhatsApp Webhooks, Razorpay Payment Callbacks, and Automated WhatsApp Reminders cleanly.

4. Storage for Lab Reports

Built-in secure object storage to hold ECG graphs, lab blood test PDF reports, and digital invoices.

Comparison: Custom Supabase Backend vs Traditional Third-Party CRM

Evaluation Metric	Supabase Built Backend (Recommended)	External Third-Party CRM
Cost Structure	✓ Very Low (\$0 to \$25/mo pay-as-you-grow)	Expensive monthly per-user licensing (\$150-\$500+/mo)
Custom Service Logic	✓ 100% Tailored for Healthcare Slots & Pricing	Rigid workflow structures; difficult to customize
WhatsApp API Integration	✓ Direct Meta API / Webhooks Control	Requires expensive middleware / third-party connectors
Realtime Dispatching	✓ Native WebSocket Realtime Engine	Delayed polling updates (30s - 2min latency)
Reusability (AutomateX Product)	✓ Turnkey SaaS platform for multiple clients	Single client lock-in

4 Admin Operations CRM Dashboard Blueprint

The Admin CRM visualizes all live bookings and staff operations via an intuitive Kanban & Pipeline interface:

📄 New Enquiries & Leads

Live log of all ongoing WhatsApp chats, incoming patient inquiries, and requested slots.

👤 Staff Allocation Hub

Matches patient pincode with nearest available Nurse, Physio, or Lab Tech for 1-click assignment.

🚚 Live Visit Tracking

Pipeline view: *Pending* → *Assigned* → *On the Way* → *In Progress* → *Completed*.

💰 Payment & Invoicing

Tracks Paid vs Pending payments, automatically dispatches Razorpay reminders & PDF receipts.

★ Ratings & Patient Feedback

Aggregates post-service star ratings, review notes, and patient retention analytics.

🚨 Emergency Escalation Log

Instant red-flag notifications for clinical emergency triage cases requiring immediate doctor calls.

Week 1: Core & DB • Supabase DB Schemas • WhatsApp Meta API Setup • Conversation State Machine	Week 2: Booking Engine • Service & Slot Engine • Pricing Calculator • Emergency Safety Triage	Week 3: Admin CRM UI • Next.js/React Dashboard • Realtime Dispatch Board • Staff Assignment Flow	Week 4: Payment & Launch • Razorpay Payment Links • PDF Invoicing & Feedback • End-to-End Testing & UAT
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